Linear Algebra Final Exam Standard Answer Key (Complete and Error-Free Version)

Problem 1: Compute the determinant of order n

$$D_n = \begin{vmatrix} a_1+\lambda & a_2 & \cdots & a_n \\ a_1 & a_2+\lambda & \cdots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & a_n+\lambda \end{vmatrix}$$

Column operations: add columns $2, 3, \ldots, n$ all to column 1; every entry of column 1 becomes $\lambda + \sum_{i=1}^{n} a_i$.

Factor out the common factor:

$$D_n = \left(\lambda + \sum_{i=1}^{n} a_i\right) \begin{vmatrix} 1 & a_2 & \cdots & a_n \\ 1 & a_2+\lambda & \cdots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_2 & \cdots & a_n+\lambda \end{vmatrix}$$

Row operations: subtract row 1 from row i ($i = 2, \ldots, n$), reducing to upper triangular form:

$$D_n = \left(\lambda + \sum_{i=1}^{n} a_i\right) \begin{vmatrix} 1 & a_2 & \cdots & a_n \\ 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda \end{vmatrix}$$

Multiply the entries on the main diagonal:

$$\boxed{D_n = \lambda^{n-1}\left(\lambda + a_1 + a_2 + \cdots + a_n\right)}$$

Problem 2: Vector operations

Let $\alpha = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$; find $\alpha^T\alpha,\ \alpha\alpha^T,\ \alpha\alpha^T\alpha,\ (\alpha\alpha^T)^{100}$.

(1) $\alpha^T\alpha$ (a scalar):

$$\alpha^T\alpha = 1^2 + 2^2 + 1^2 = \mathbf{6}$$

(2)

$$\alpha\alpha^T = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

(3)

$$\alpha\alpha^T\alpha = \alpha \cdot (\alpha^T\alpha) = 6\alpha = \begin{pmatrix} 6 \\ 12 \\ 6 \end{pmatrix}$$

(4)

$$B^2 = (\alpha\alpha^T)(\alpha\alpha^T) = \alpha(\alpha^T\alpha)\alpha^T = 6\alpha\alpha^T = 6B$$

Induction gives $B^n = 6^{n-1}B$ ($n \geq 1$), hence

$$\boxed{(\alpha\alpha^T)^{100} = 6^{99}\alpha\alpha^T = 6^{99}\begin{pmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{pmatrix}}$$

Problem 3: Prove that $A - E$ is invertible and find its inverse

We are given that the square matrix A of order n satisfies $A^2 - 5A + E = 0$.

Proof: apply algebraic rearrangement to the given equation:

$$\begin{aligned} A^2 - 5A + E &= 0\\ A^2 - A - 4A + 4E &= 3E\\ (A - E)(A - 4E) &= 3E\\ (A - E)\cdot\frac{1}{3}(A - 4E) &= E \end{aligned}$$

By the definition of an invertible matrix, $A - E$ is invertible, and

$$\boxed{(A - E)^{-1} = \frac{1}{3}(A - 4E)}$$

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Problem 4: Discuss the solutions of the system of linear equations

$$\begin{cases} ax_1 + x_2 + x_3 = 0\\ x_1 + ax_2 + x_3 = 2\\ x_1 + x_2 + ax_3 = a \end{cases}$$

Compute the determinant of the coefficient matrix:

$$|A| = \begin{vmatrix} a & 1 & 1\\ 1 & a & 1\\ 1 & 1 & a \end{vmatrix} = (a + 2)(a - 1)^2$$

(1)

(2) **No solution ($a = 1$)**: the augmented matrix

$$\bar{A} = \begin{pmatrix}1 & 1 & 1 & 0\\ 1 & 1 & 1 & 2\\ 1 & 1 & 1 & 1\end{pmatrix} \to \begin{pmatrix}1 & 1 & 1 & 0\\ 0 & 0 & 0 & 2\\ 0 & 0 & 0 & 1\end{pmatrix}$$

$r(A) = 1 \neq r(\bar{A}) = 2$, so there is no solution when $\mathbf{a = 1}$.

(3) **Infinitely many solutions ($a = -2$)**: apply row operations to the augmented matrix:

$$\bar{A} = \begin{pmatrix}-2 & 1 & 1 & 0\\ 1 & -2 & 1 & 2\\ 1 & 1 & -2 & -2\end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix}1 & -2 & 1 & 2\\ -2 & 1 & 1 & 0\\ 1 & 1 & -2 & -2\end{pmatrix} \xrightarrow{R_2 + 2R_1,\ R_3 - R_1} \begin{pmatrix}1 & -2 & 1 & 2\\ 0 & -3 & 3 & 4\\ 0 & 3 & -3 & -4\end{pmatrix} \xrightarrow{R_3 + R_2} \begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$$

$r(A) = r(\bar{A}) = 2 < 3$, so the system has infinitely many solutions. The equivalent system is:

$$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ -3x_2 + 3x_3 = 4 \end{cases}$$

Let the free variable be $x_3 = c$; then

$$x_2 = c - \frac{4}{3}, \quad x_1 = c - \frac{2}{3}$$

$$\boxed{\text{General solution:} \quad X = c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ -\frac{4}{3} \\ 0 \end{pmatrix}, \quad c \in \mathbb{R}}$$

Problem 5: Prove that $\beta_1, \beta_2, \beta_3$ are linearly independent

Given that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent, and $\beta_1 = \alpha_1 + \alpha_2$, $\beta_2 = \alpha_2 + \alpha_3$, $\beta_3 = \alpha_3 + \alpha_1$.

Proof: set $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$; substitute the expressions of the β_i and collect terms:

$$(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$$

Since $\alpha_1, \alpha_2, \alpha_3$ are linearly independent, every coefficient must be zero:

$$\begin{cases} k_1 + k_3 = 0 \\ k_1 + k_2 = 0 \\ k_2 + k_3 = 0 \end{cases}$$

The determinant of the coefficients of this homogeneous system is

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 1 \cdot (1 - 0) - 0 + 1 \cdot (1 - 0) = 2 \neq 0$$

Hence the system has only the zero solution $k_1 = k_2 = k_3 = 0$, and therefore $\beta_1, \beta_2, \beta_3$ are linearly independent.

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Problem 6: Determine the parameters and find the maximal linearly independent subset

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & a \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & b \end{pmatrix}$$

Elementary row operations $(R_2 - 3R_1,\ R_4 - 5R_1)$:

$$\to \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -6 & a - 3 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & -1 & -2 & -6 & b - 5 \end{pmatrix}$$

Continue $(R_2 + R_3,\ R_4 + R_3)$:

$$\to \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & a \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & b-2 \end{pmatrix}$$

Swap ($R_2 \leftrightarrow R_3$) and arrange into row echelon form:

$$\to \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & a \\ 0 & 0 & 0 & 0 & b-2 \end{pmatrix}$$

Determine the parameters: for $r(A) = 2$, rows 3 and 4 must be all-zero rows, hence

$$a = 0, \quad b - 2 = 0 \implies \mathbf{a = 0},\ \mathbf{b = 2}$$

Find the maximal linearly independent subset (substitute $a = 0,\ b = 2$ and continue simplifying with $R_1 - R_2$):

$$A \to \begin{pmatrix} 1 & 0 & -1 & -5 & -2 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The pivots lie in columns 1 and 2, so a **maximal linearly independent subset** of the set of column vectors is $\{\alpha_1, \alpha_2\}$.

Linear expressions of the remaining column vectors (read off directly from the corresponding columns of the

Problem 7: Find the eigenvalues and eigenvectors

(1) $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Characteristic equation:

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & -1 & 0 \\ -1 & \lambda - 1 & 0 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = (\lambda - 1)\left[(\lambda - 1)^2 - 1\right] = \lambda(\lambda - 1)(\lambda - 2) = 0$$

Eigenvalues: $\lambda_1 = 0,\ \lambda_2 = 1,\ \lambda_3 = 2$

$\lambda_1 = 0$: solve $-Ax = 0$

$$(-A) = \begin{pmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies x_1 + x_2 = 0,\ x_3 = 0$$

Taking $x_2 = 1$ gives the fundamental system of solutions $\xi_1 = (-1,\ 1,\ 0)^T$, and the corresponding eigenvector is $k_1\xi_1$ ($k_1 \neq 0$).

$\lambda_2 = 1$: solve $(E - A)x = 0$

$$E - A = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies x_1 = 0,\ x_2 = 0$$

Taking $x_3 = 1$ gives the fundamental system of solutions $\xi_2 = (0,\ 0,\ 1)^T$, and the corresponding eigenvector is $k_2\xi_2$ ($k_2 \neq 0$).

$\lambda_3 = 2$: solve $(2E - A)x = 0$

$$2E - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies x_1 - x_2 = 0,\ x_3 = 0$$

Taking $x_1 = 1$ gives the fundamental system of solutions $\xi_3 = (1,\ 1,\ 0)^T$, and the corresponding eigenvector is $k_3\xi_3$ ($k_3 \neq 0$).

(2) $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

Characteristic equation:

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & -1 \\ -1 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 1 = \lambda(\lambda - 2) = 0$$

Eigenvalues: $\lambda_1 = 0,\ \lambda_2 = 2$

$\lambda_1 = 0$: solve $-Ax = 0$

$$-A = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \implies x_1 + x_2 = 0$$

Taking $x_2 = 1$ gives the fundamental system of solutions $\xi_1 = (-1,\ 1)^T$, and the corresponding eigenvector is $k_1\xi_1$ ($k_1 \neq 0$).

$\lambda_2 = 2$: solve $(2E - A)x = 0$

$$2E - A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \implies x_1 - x_2 = 0$$

Taking $x_2 = 1$ gives the fundamental system of solutions $\xi_2 = (1,\ 1)^T$, and the corresponding eigenvector is $k_2\xi_2$ ($k_2 \neq 0$).